Let’s look at the earlier example again. When committing to $1 per hour for one year of `m5` instances running in `us-east-1` and paying $8,760.00 upfront, you will get an `m5.large` EC2 instance at a discount of 42% in US East (N. Virginia). Compare that to the 31% discount when purchasing a Compute Savings Plan instead.

WARNING Buying a reservation will incur costs for one or three years. That’s why we did not add an example for this section.

Think of Savings Plans as a way to optimize your AWS bill. Buying a Savings Plan does not have any effect on your running EC2 instances. Also, an on-demand instance gets billed under the conditions of a Savings Plan automatically. There is no need to restart or modify an EC2 instance.

Capacity Reservations

As mentioned, Savings Plans do not have any effect on EC2 instances, only on billing. In contrast, EC2 Capacity Reservations allow you to reserve capacities in AWS’s data centers. A Capacity Reservation comes with an hourly fee. For example, when you reserve the capacity for an `m5.large` instance, you will pay the typical on-demand fee of $0.096 per hour, no matter whether or not an EC2 instance of that type is running. In return, AWS guarantees the capacity to launch an `m5.large` instance at anytime.

With on-demand instances, you might run into the case that AWS cannot fulfill your request of spinning up a virtual machine. This type of situation might happen for rare instance types, during peak hours, or in special cases like outages causing many AWS customers to replace their failed instances.

If you need to guarantee that you are able to spin up an EC2 instance at anytime, consider EC2 Capacity Reservations.

In summary, we highly recommend purchasing Savings Plans for workloads, where predicting the resource consumption for the next year is possible. It is worth noting that it is not necessary to cover 100% of your